nature communications

Article

https://doi.org/10.1038/s41467-025-67981-1

Excess mortality in Europe estimated by EuroMOMO during the COVID-19 pandemic and previous influenza seasons

Received: 17 December 2024

Accepted: 14 December 2025

Published online: 15 January 2026

Check for updates

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Important questions remain regarding differences in geographical and age-specific mortality patterns as the COVID-19 pandemic evolved in consecutive waves, and how COVID-19 mortality compares to seasonal influenza. In a pandemic situation, excess all-cause mortality provides a more complete and robust measure than cause-specific mortality. Data submitted by 26 countries participating in the European Mortality Monitoring (EuroMOMO) network between 2020 and 2023 was analysed to quantify excess all-cause mortality during the COVID-19 pandemic. Excess mortality from this period was compared to previous influenza seasons from 2014 to 2019. Pooled estimates of excess mortality showed four main waves during the COVID-19 pandemic period, most markedly in people aged 65 years and above, with timing and magnitude that varied between countries. Here we show that prior to implementation of control measures and COVID-19 vaccination, excess mortality greatly exceeded typical seasonal influenza mortality, but later during the pandemic was at levels comparable to influenza.

Following its emergence in late 2019, the novel severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) and the associated coronavirus disease 2019 (COVID-19) rapidly evolved into a global pandemic, which has had dramatic consequences for public health and claimed millions of lives worldwide (20200121-sitrep-1-2019-ncov.pdf.)[1–4]. In Europe, COVID-19 emerged in early 2020, after a winter season with relatively low levels of seasonal influenza in most countries[5], and from then spread geographically and over time in

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consecutive epidemic waves across the continent. The timing of the epidemic waves and the associated mortality impact of COVID-19 were influenced by the timing and extent of the public health and social measures (PHSM) that were implemented by governments, the emergence of new SARS-CoV-2 variants of concern, and the level of naturally acquired and vaccine-induced population immunity[6–9].

During the initial years of the pandemic, i.e., from early 2020 to the middle of 2023, Europe experienced four main epidemic waves of COVID-19, as documented by the numbers of confirmed COVID-19 cases and deaths notified by countries[10,11]. However, the confirmed COVID-19 deaths only represent a part of the actual mortality burden of the pandemic, dependent on availability and practices of COVID-19 testing and reporting, and do not capture the indirect mortality associated with the COVID-19 pandemic, linked to inadequate health seeking for other illnesses and overburdened health services in many places.

Estimating excess all-cause mortality is considered to provide a more complete and objective measure of the full mortality impact of the COVID-19 pandemic in the population[12]. Monitoring excess mortality during influenza seasons has been practiced routinely for several years by the European Mortality Monitoring Network (EuroMOMO), established during the 2009 influenza pandemic and using excess mortality as a means of estimating the full influenza mortality burden, capturing also the hidden burden of untested influenza cases and deaths and hence serving as a more complete measure of the impact of the dominant circulating influenza virus subtype on mortality in any given season[13,14]. The purpose and scope of EuroMOMO is to detect and quantify all-cause excess mortality temporarily associated with seasonal influenza, new pandemics, and other acute public health threats in a timely manner to rapidly inform public health policy and practice[15]. The monitoring results are published in weekly bulletins on the EuroMOMO website[16]. Therefore, from the very onset of the COVID-19 pandemic in Europe in February 2020, the EuroMOMO network monitored the excess mortality burden and reported this weekly to inform policy makers and the public. EuroMOMO reports and graphs were widely cited by the media worldwide[17,18] and used to understand the impact of the pandemic as it evolved. Some preliminary estimates of excess mortality during the first wave of the COVID-19 pandemic were published by EuroMOMO in July 2020[19], which documented the steep mortality peak following the first wave of COVID-19 in Europe from February 2020, and again during the second wave in late 2020 and early 2021[20]. As the pandemic continued, other organizations similarly embarked on excess mortality monitoring to understand the full deaths toll of COVID-19, applying varying sources of data and statistical approaches, sometimes with differing results[21–26].

As the COVID-19 pandemic from May 2023 was no longer considered a public health emergency of international concern, as declared by WHO[27], we set out to better understand the mortality burden associated with the pandemic period in Europe, from week 8/2020 to week 20/2023, compared to the pre-pandemic period. We applied our standard EuroMOMO methodology to analyze country-specific and European-wide pooled excess mortality estimates for the 26 participating countries during the COVID-19 pandemic period and during each of the observed main waves of excess mortality in this period. Furthermore, for a subset of 16 countries, we compared the excess mortality during the COVID-19 pandemic to the mortality estimated by EuroMOMO in the five previous influenza seasons, from the 2014/15 season to the 2018/19 season, as a means to assess in a systematic way the relative mortality burden of COVID-19 versus seasonal influenza, which became a topic of much attention as the COVID-19 pandemic evolved[28].

Results

Pooled excess mortality during the COVID-19 pandemic period

Overall during the entire COVID-19 pandemic period, from week 8/2020 to week 20/2023, a total pooled all-cause excess mortality rate of 91.7 deaths per 100,000 pyrs (95% CI: 91.0; 92.4) was observed in the 26 participating countries (Table 1), corresponding to 9.5% higher mortality than expected. Excess mortality was primarily observed in individuals aged 65 years and above, with the highest estimated excess mortality in elderly aged 85 years and above at 1390 (95% CI: 1377; 1404) excess deaths per 100,000 pyrs (Supplemental Table 1), or 10.0% higher than expected.

The temporal change in weekly pooled excess mortality rates for all ages across the full study period are shown in Fig. 1A, while age-stratified mortality rates are shown in Fig. 1B. The rates of observed and expected deaths are shown in Supplemental Fig. 2.

Pooled excess mortality during four main waves of COVID-19

During the entire pandemic period, several temporal peaks of excess all-cause mortality were observed. Apart from the total pandemic period, spanning week 8/2020 to week 20/2023, four distinct intervals emerged as having the longest durations of excess mortality (Table 1). Combined, these four main waves represent 75% of the total study period. The first main wave was the shortest, lasting 12 weeks compared to 25, 47, and 43 weeks for the second, third, and fourth main wave, respectively.

The first wave of excess mortality among all ages was observed from week 10/2020 up to and including week 21/2020, and had a mortality rate of 199.5 (95% CI: 196.3; 202.7) pooled all-cause excess deaths per 100,000 pyrs, corresponding to 20.5% above the expected level. The pooled excess mortality increased with age, and the highest number of excess deaths over the period was observed among those aged 85 years and older, with 3561.5 (95% CI: 3492.2; 3631.3) excess deaths per 100,000 pyrs (Table 1), or 25.0% above the expected. The second main wave of pooled excess mortality among all ages was observed from week 37/2020 up to and including week 8/2021 and had 173.8 (95% CI: 171.5; 176.0) excess deaths per 100,000 pyrs, which corresponds to 17.5% above the expected level. The third main wave lasted from week 26/2021 to week 20/2022 and had a pooled excess mortality rate of 86.4 (95% CI: 85.4; 87.7) excess deaths per 100,000 pyrs, equivalent to 8.9% above the expected number of deaths. Lastly, the fourth main wave lasted from week 22/2022, to week 12/2023 and had a cumulative excess mortality rate of 97.6 (95% CI: 96.2; 99.0) excess deaths per 100,000 pyrs, or 10.0% above the expected level.

The pooled excess mortality rates by age group during the four main waves are shown in Fig. 2. The highest excess mortality was observed in the first and second main waves among the oldest age groups.

The excess mortality observed by the EuroMOMO network during past influenza periods, from the 2014/2015 season to the 2018/19 season, in comparison to the four main excess waves during the COVID-19 pandemic, is shown in Fig. 3. This comparison includes data from the 16 countries that consistently provided data throughout all the previous influenza seasons. The pooled excess mortality (deaths per 100,000 pyrs) during the recent influenza periods ranged from 31.9 (95% CI: 30.0; 33.9) in the 2015/16 season to 120.4 (95% CI: 117.8; 123.0) in the 2017/18 season, see Table 2. In contrast, the excess mortality observed during the four main excess waves of the COVID-19 pandemic among the 16 countries with consistent data were: 258.8 (95% CI: 254.9; 262.8) in wave 1, 183.4 (95% CI: 180.9; 186.0) in wave 2, 88.7 (95% CI: 87.3; 90.1) in wave 3, and 87.2 (95% CI: 85.3; 88.7) in wave 4. Notably, the levels for the main third and fourth waves were similar to the periods of excess mortality during previous influenza seasons. Similar results were found when all countries in the network were included in the analyses, where data were available, see Supplemental Table 2 and Supplemental Fig. 1.

Country-level excess all-cause mortality

Important differences in timing and maximum levels of weekly excess mortality were observed across the participating European countries

https://doi.org/10.1038/s41467-025-67981-1

Table 1 | Pooled estimates of excess all-cause mortality over the entire study period and by main waves of excess mortality, for all ages and among 65 years and above

Age group	Period	100,000 person-years observed n	Observed all-cause deaths, No. n	Expected all-cause deaths, No. n	Excess all-cause deaths, No. n [95%- CI]	Excess all-cause deaths per 100,000 person-years n [95%- CI]
All ages	Study period (week 8/2020 to week 20/2023)	12698.5	13474922	12310223	1164699 (1156061;1173357)	91.7 (91.0;92.4)
All ages	Wave 1 (week 10/2020 to week 21/2020)	892.4	1046613	868584	178029 (175148;180925)	199.5 (196.3;202.7)
All ages	Wave 2 (week 37/2020 to week 8/2021)	1861.2	2170794	1847361	323433 (319282;327602)	173.8 (171.5;176.0)
All ages	Wave 3 (week 26/2021 to week 20/2022)	3508.0	3726010	3422899	303111 (298639;307605)	86.4 (85.1;87.7)
All ages	Wave 4 (week 22/2022 to week 12/2023)	3228.6	3459180	3144152	315028 (310490;319588)	97.6 (96.2;99.0)
≥65 years	Study period (week 8/2020 to week 20/2023)	2576.2	11571214	10546191	1025023 (1016959;1033109)	397.9 (394.7;401.0)
≥65 years	Wave 1 (week 10/2020 to week 21/2020)	178.5	907262	742810	164452 (161721;167199)	921.4 (906.1;936.8)
≥65 years	Wave 2 (week 37/2020 to week 8/2021)	374.6	1877273	1584679	292594 (288680;296524)	781.1 (770.6;791.6)
≥65 years	Wave 3 (week 26/2021 to week 20/2022)	712.9	3192524	2935220	257304 (253178;261451)	360.9 (355.2;366.8)
≥65 years	Wave 4 (week 22/2022 to week 12/2023)	660.6	2982804	2700863	281941 (277677;286226)	426.8 (420.3;433.3)

during the COVID-19 pandemic. Some countries experienced peaks of high levels of mortality, whereas other countries experienced only modest excess mortality, as shown in Fig. 4. Although the majority of countries experienced the most dramatic levels of excess mortality during the first wave of COVID-19, in early 2020, some countries were more severely affected during the second or third wave, as shown in Fig. 5.

The countries that experienced the highest excess mortality during the first wave were Belgium, England (UK), France, Italy, Netherlands, Northern Ireland (UK), Scotland (UK), Spain, Sweden, and Wales (UK). During the second wave, the countries that experienced the highest excess mortality were Austria, Belgium, France, Germany, Greece, Hungary, Italy, Malta, Netherlands, Portugal, Slovenia, and Spain. During the third wave, mainly Cyprus, Estonia, Germany, Greece, Hungary, Italy, Malta, and Slovenia experienced excess mortality, while Germany, Greece, Italy, Portugal, and Spain experienced most excess mortality during the fourth wave (Fig. 5).

Discussion

When the COVID-19 pandemic first emerged, its severity and potential impact on population mortality was unknown. The estimates of excess all-cause mortality presented here reflect how the EuroMOMO network detected changing levels of mortality as the COVID-19 pandemic evolved week by week across Europe, as a proxy measure for the pandemic's total direct and indirect mortality burden in different age groups. With the retrospective analysis presented here, we aimed to provide a more in-depth understanding of the temporal changes in excess mortality observed over the course of the COVID-19 pandemic, with a particular focus on the consecutive waves of excess mortality, and to assess the pandemic's mortality impact relative to recent epidemics of seasonal influenza.

We found overall a 9.5% higher mortality than expected over the pandemic period. This excess mortality burden was particularly carried by the elderly, with a 10.0% higher mortality among people aged 85 years and above, while no or only modest excess mortality was detected in children and younger adults below 45 years of age. However, it should be noted that the estimates for individuals aged 0–14 years are more challenging to fit in the model, as illustrated in Supplemental Fig. 2, and should be interpreted with some caution.

The COVID-19 excess mortality burden showed wide variation over time and place over the course of the pandemic, forming four more or less distinct temporal waves, which reflect the combined effect of shifts into new dominating SARS-CoV-2 virus variants, the impact of various PHSM countermeasures, and build-up of population and vaccine-induced immunity with variable uptake for the COVID-19 vaccine campaigns.

The first large mortality wave, in early 2020, was caused by the Wuhan variant, followed by consecutive waves linked to the subsequent Alpha, Delta, and Omicron variants, respectively, with partial overlaps between them and with some variation between countries[29]. The four waves showed an overall declining mortality impact over time, although with important differences between the countries, reflecting the degree of community transmission of SARS-CoV-2, the nature of different PHSMs applied by countries to different extent and timing, introduction and uptake of COVID-19 vaccination in priority groups, and multiple other societal and demographic factors[30–34]. These factors all varied between countries, resulting in different mortality impacts. Furthermore, other respiratory infections have contributed to mortality levels to a varying extent during the pandemic, with their circulation in society being markedly impacted by the implementation and easing of PHSMs during different phases of the pandemic[35]. During the first wave in 2020, little influenza circulated in most European countries, and only returned in 2022 after the lifting of the many restrictions implemented initially and after COVID-19 vaccines had been introduced. Thus, the third

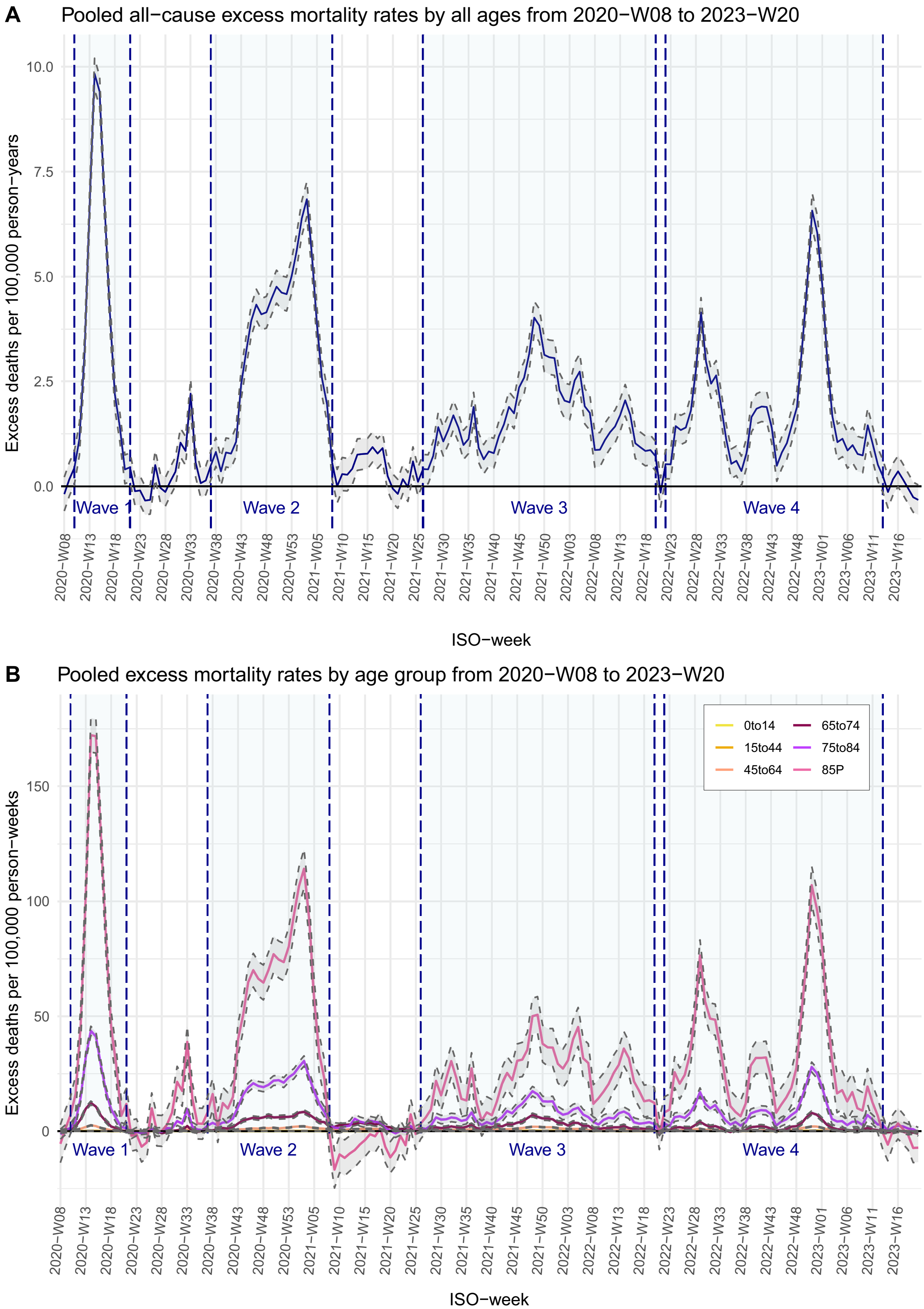

Fig. 1 | Weekly pooled excess mortality rates and the four main waves. A all ages and **B** by age groups. The panels show the pooled weekly excess all-cause mortality rates per 100,000 person-weeks with 95% confidence intervals. This means that data are shown as the total estimated excess deaths per 100,000 person-weeks (blue line) +/− the 95% confidence interval (shaded grey area). The excess periods respectively cover week 10/2020 to 21/2020, week 37/2020 to 8/2021, week 26/2021 to 20/2022, and week 22/2022 to 12/2023 for main waves 1, 2, 3, and 4, which are highlighted with light blue shading. **A** shows results for all ages, and **B** presents results by age group, with age categories displayed in different colors: 0–14 years (yellow), 15–44 years (orange), 45–64 years (salmon), 65–74 years (brown), 75–84 years (purple), and 85 P years (representing individuals aged 85 years and above) (pink). Please note that the confidence intervals are drawn vertically and therefore appear very narrow where the curve is steep.

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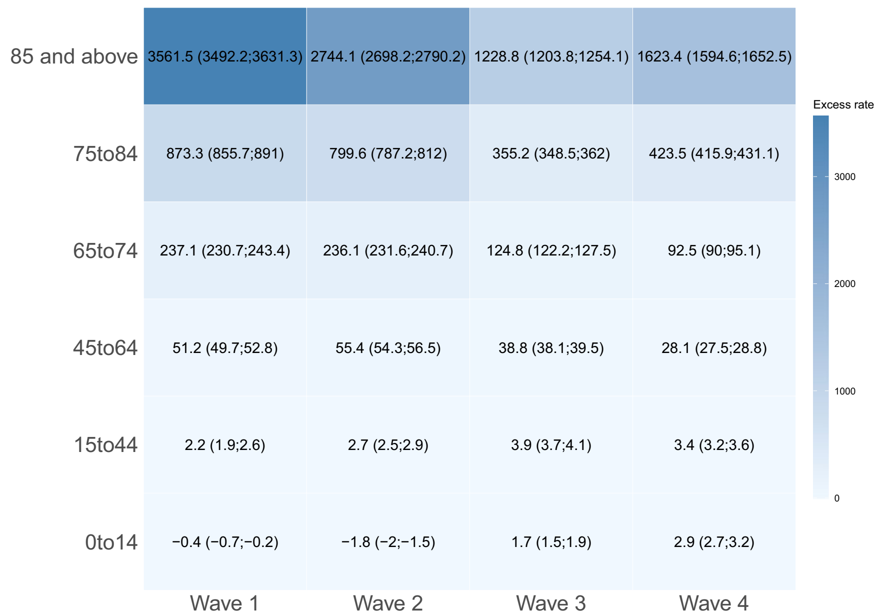

Fig. 2 | Heatmap of pooled excess mortality rates per 100,000 person-years by age group for the four main waves. Figure 2 shows the pooled excess all-cause mortality rates per 100,00 person-years with 95% confidence intervals by main wave and age group. The color scale represents excess mortality per 100,000 person-years. Low values are shown in white, while increasing excess mortality estimates are displayed in increasingly darker blue. The periods of excess mortality cover week 10/2020 to 21/2020, week 37/2020 to 8/2021, week 26/2021 to 20/2022, and week 22/2022 to 12/2023 for the main excess waves 1, 2, 3, and 4, respectively.

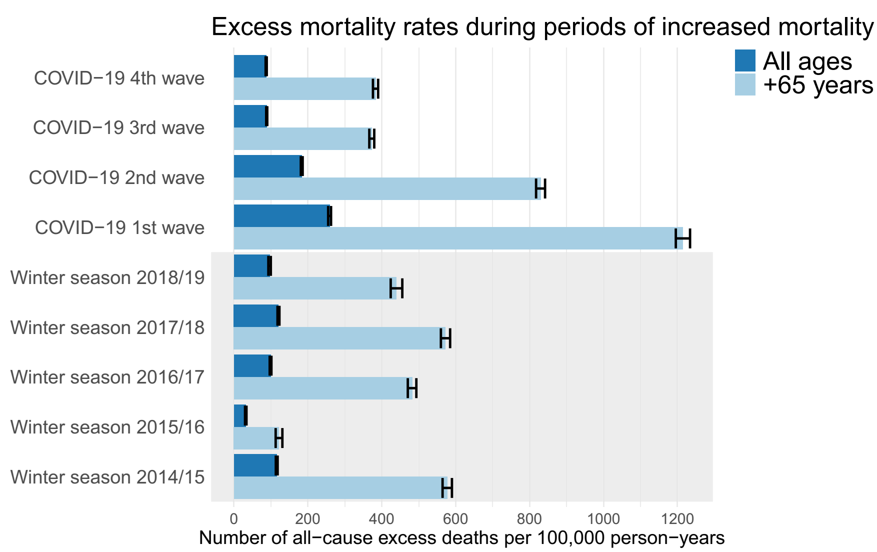

Fig. 3 | All-cause excess deaths per 100,000 person-years during the winter excess waves from 2014/15 to 2018/19, combined with the four main excess waves during the COVID-19 pandemic (among the 16 countries with data for the entire period). This Figure shows the pooled estimated excess deaths per 100,000 person-years with 95% confidence intervals for the winter periods of excess mortality from 2014/15 to 2018/19 together with the four main waves of excess mortality during the COVID-19 pandemic for all ages and for the age group of 65 years and above. This means that the center of each error bar represents the total estimated excess deaths per 100,000 person-years, and the length of the error bar indicates the 95% confidence interval. The dark blue bars represent the estimates for all ages, while the light blue bars represent the estimates for individuals aged 65 years and above. The winter season periods are highlighted with light grey shading. The periods of excess mortality cover week 10/2020 to 21/2020, week 37/2020 to 8/2021, week 26/2021 to 20/2022, and week 22/2022 to 12/2023 for the main excess waves 1, 2, 3 and 4, respectively, during the COVID-19 pandemic. The periods of excess mortality for the influenza seasons 2014/2015, 2015/16, 2016/17, 2017/18 and 2018/19, respectively, span from week 49/2014 to 15/2015, week 1/2016 to 13/2016, week 41/2016 to 8/2017, week 47/2017 to 15/2018, and week 51/2018 to week 15/2019. The 16 included countries are: Belgium, Denmark, Estonia, Finland, France, Greece, Hungary, Ireland, Netherlands, Norway, Portugal, Spain, Switzerland, UK (England), UK (Scotland), and UK (Wales).

and particularly the fourth wave of excess mortality is partially explained by the return of the common seasonal respiratory infections, such as influenza and RSV in addition to COVID-19. Finally, another important factor influencing excess mortality is the effect of temperature, i.e., cold snaps during winter and heat waves during the summer, which will have coincided and contributed to varying degrees to the excess mortality during the different waves of COVID-19 across the participating countries[36,37].

The highest cumulated pooled all-cause excess mortality was observed in the first main wave from week 8 to week 21, 2020, with a

https://doi.org/10.1038/s41467-025-67981-1

Table 2 | Excess all-cause number of death rates per 100,000 person-years among the 16 countries with data for the entire period, for the winter seasons from 2014/15 to 2018/19

Season	2014/15 week 49 to week 15	2015/16 week 1 to week 13	2016/17 week 41 to week 8	2017/18 week 47 to week 15	2018/19 week 51 to week 10
Dominant type of influenza	A(H3N2)	A(H1N1)pdm09 + B/Victoria	A(H3N2)	B/Yamagata + mixed A	Mixed A
WHO recommended vaccine	A(H3N2) A(H1N1)pdm09 B/Yamagata	A(H3N2) A(H1N1) pdm09 B/Yamagata	A(H3N2) A(H1N1) pdm09 B/Victoria	A(H3N2) A(H1N1)pdm09 B/Victoria	A(H3N2) A(H1N1) pdm09 B/Victoria
Age groups	Excess all-cause mortality per 100,000 person-years (95% CI)				
≥65	577.09 (564.5;589.8)	121.76 (112.8;131.0)	481.68 (470.2;493.3)	572.14 (559.7;584.7)	426.27 (411.2;441.6)
Total	115.17 (113.2;118.2)	31.91 (30.0;33.9)	98.99 (96.7;101.3)	120.41 (117.8;123.0)	92.21 (89.0;95.4)
Number of countries participating	16[a]	16[a]	16[a]	16[a]	16[a]
Population covered (millions)	250	250	250	250	250

[a]Belgium, Denmark, England (UK), Estonia, Finland, France, Greece, Hungary, Ireland, Netherlands, Norway, Portugal, Scotland (UK), Spain, Switzerland, and Wales (UK).

mortality rate of 199.5 (95% CI: 196.3; 202.7) excess deaths per 100,000 pyrs, or 20.5% higher than expected. This overall figure is at a magnitude that is rarely seen in an ordinary influenza season, and it does, furthermore, not reflect the large variation seen across countries; a variation that most likely can be ascribed to the different timing and extent of viral spread and countermeasures. As the circulation of other respiratory infections had been widely excluded by the implementation of the PHSMs, and in the absence of any naturally acquired or vaccine-induced protective immunity in the population, the excess mortality seen in the first wave can largely be attributed to the effect of COVID-19 alone. The observed high mortality rates experienced in many countries, despite the dramatic reduction in mortality impact brought about by the large-scale public health response, show just how serious the pandemic potential was.

Importantly, there were marked differences in country-level estimated excess mortality, varying both between the participating countries and between the main waves within countries (Fig. 4). National responses (both therapeutic and PHSM) to the pandemic varied vastly, which consequently affected the spread of disease by country, which again was impacted by differences in healthcare services, COVID-19 vaccine roll-out and uptake, differences in public health recommendations and implementation of non-pharmaceutical measures, e.g., the degree and timing of societal lockdowns. Therefore, the waves of pooled excess all-cause mortality do not necessarily align with the waves of excess mortality observed by individual countries.

When we compared the excess mortality during the COVID-19 pandemic to the five previous influenza seasons, we found that COVID-19 in the first and second main wave had a considerably higher mortality impact compared to the pre-pandemic winters. The observed mortality during the first excess mortality wave during the COVID-19 pandemic was about twofold higher than the worst seasons of influenza, in 2014/15 and 2017/18, and many times higher than the milder seasons. Without the rapid implementation of PHSMs during the first COVID-19 wave, the magnitude of difference would likely have been greater. It should also be considered that it is not an all-things-being-equal-comparison, for example, because the global population was completely naïve to COVID-19. In a counterfactual scenario with high uptake of influenza vaccines in eligible populations and restrictions in society, it is likely that the previous influenza seasons would have been milder.

The higher estimates of excess mortality during the COVID-19 pandemic compared to seasonal influenza at the start of the pandemic is well in line with clinical studies pointing to a higher severity of COVID-19 combined with the population's naivety to the virus. The lower excess mortality observed in the third and fourth waves, similar to the historic influenza seasons, is likely due to the combined effects of large scale COVID-19 vaccine roll-out, increased immunity from prior COVID-19 infections, and lower severity of the Omicron variants[38].

However, there are important caveats when comparing the mortality of the various excess mortality waves during the COVID-19 pandemic to past waves of seasonal influenza. All countries taking part in this analysis applied a number of measures to limit the spread of SARS-CoV-2 and minimize the impact of the pandemic, in particular for the elderly. Furthermore, from early 2021, most countries were successful in gradually achieving high vaccination uptake in large parts of the population. Although the timing and extent of countermeasures varied considerably, they were at much larger scale than the countermeasures usually taken against seasonal influenza, including the level of vaccination uptake and the effectiveness of the vaccines. Furthermore, seasonal influenza is being transmitted at a time of the year when other pathogens are co-circulating, whereas during the COVID-19 pandemic co-circulation of respiratory tract infections decreased as a result of the non-pharmaceutical measures applied to reduce the impact of the pandemic, and the population was immunologically naive to COVID-19, unlike to seasonal influenza. Therefore, the different waves cannot be directly compared as such and should be interpreted in the context of the mitigation strategies.

EuroMOMO uses a well-established model to estimate the expected mortality and, consequently, excess mortality. This ensures consistent monitoring and analysis of the observed mortality. However, in the COVID-19 pandemic situation, it became necessary to modify this method due to the disruption of the typical seasonal mortality pattern. As a result, the projection of the trend component in the model used for estimations was extended too far, leading to artificially elevated estimates of excess mortality in certain age groups, particularly among children of 0–14 years. Therefore, data from spring 2023 were included in our analyses to help correct the undesirable effects of an overly extended linear trend projection in the expected mortality model, which may otherwise be biased due to subsequent changes in population composition. Hence, the excess estimates for the children aged 0–14 years may in particular be less robust and potentially may benefit from population adjustment. Additionally, the baseline expected mortality may have been affected by interventions during the pandemic, such as lockdowns, which are inherently challenging to model. Moreover, it cannot be excluded that the expected mortality may have been overestimated, as the assumption of seasonal variation might have been violated due to the decline in other respiratory pathogens during the early phase of the COVID-19 pandemic, potentially leading to an underestimation of the COVID-19 effect. It should also be noted that the estimates for all ages are not adjusted for age, as this is not done in the weekly EuroMOMO monitoring. Therefore, between-country comparisons

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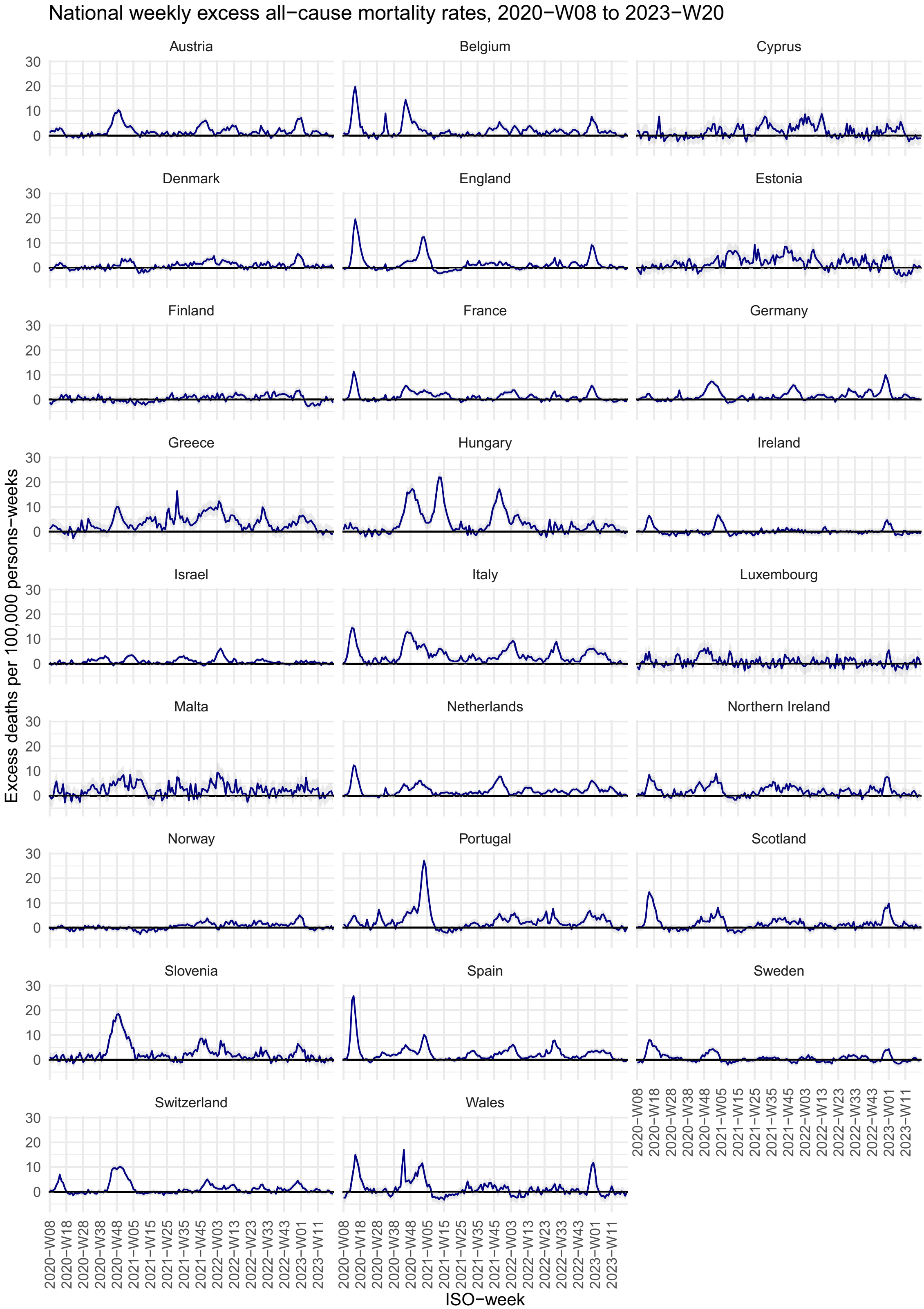

Fig. 4 | Weekly number of excess all-cause deaths as rates per 100,000 person-weeks by country. This Figure shows the national and subnational weekly excess mortality rates per 100,000 person-weeks with 95% confidence intervals from week 8, 2020, to week 20, 2023. This means that data are shown as the total estimated excess deaths per 100,000 person-weeks (blue line) +/− the 95% confidence interval (shaded grey area).

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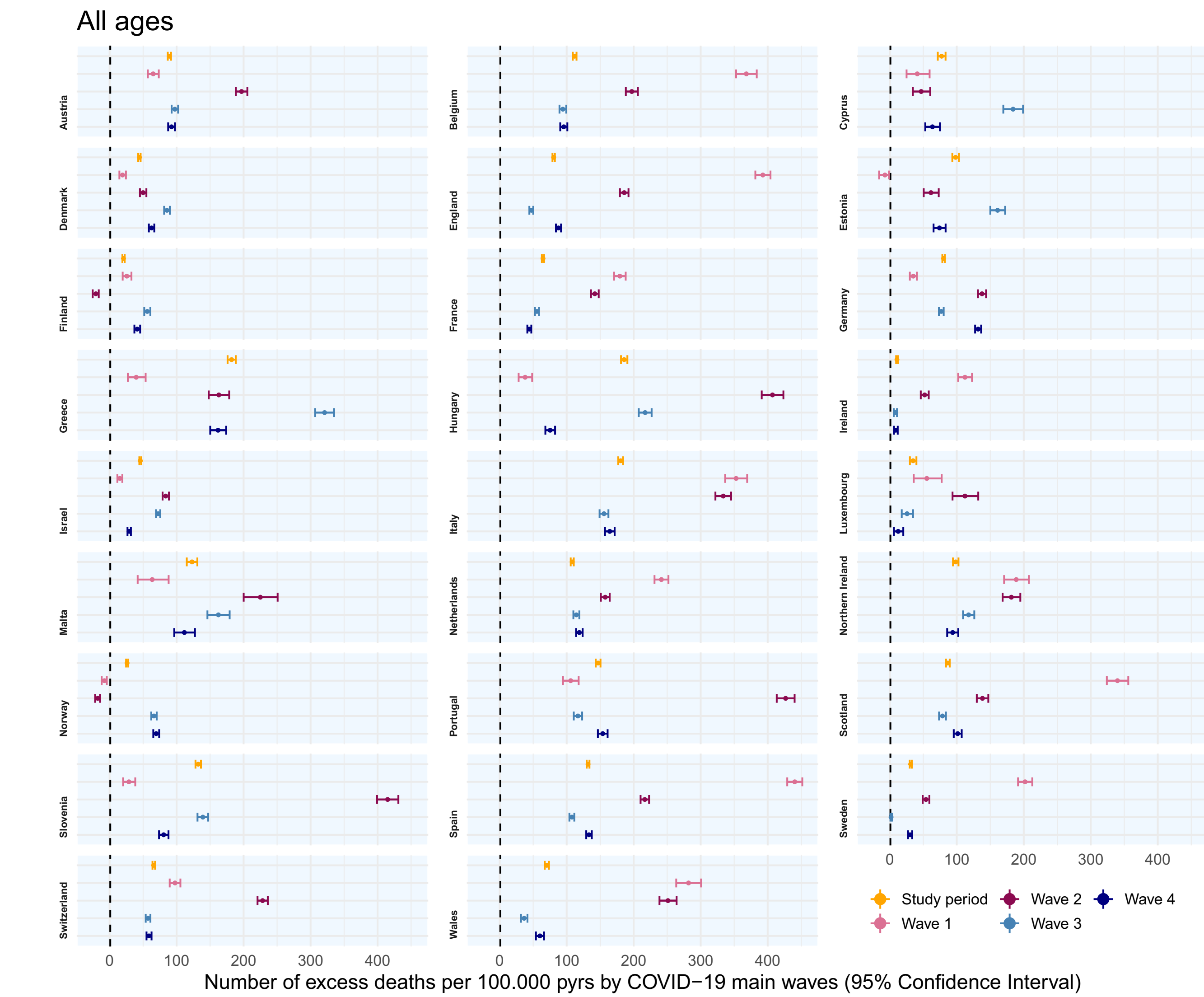

Fig. 5 | National excess all-cause number of deaths per 100,000 person-years (all ages) shown by main wave and for the entire study period for each country. This Figure shows the national and subnational excess all-cause number of deaths as rates per 100,000 person-years with 95% confidence intervals by main excess wave and for the entire study period in each country. This means that the center of each error bar represents the total estimated excess deaths per 100,000 person-years, and the length of the error bar indicates the 95% confidence interval. The excess periods respectively cover week 10/2020 to 21/2020 (pink), week 37/2020 to 8/2021 (purple), week 26/2021 to 20/2022 (light blue), and week 22/2022 to 12/2023 (dark blue) for main waves 1, 2, 3, and 4.

should be based on specific age groups, such as those aged 65 years and above. Furthermore, it should be noted that any bias or error in baseline estimation is amplified when cumulating excess mortality over time periods. Lastly, the EuroMOMO monitoring relies on the actual date of death, giving a precise view of mortality patterns compared to the registration date.

Despite these mentioned limitations, the weekly mortality estimates provided by EuroMOMO during the COVID-19 pandemic, and our retrospective analysis of COVID-19 versus seasonal influenza presented above, adds to the overall understanding of these epidemics. It should be noted that, despite some differences in applied methodology and source of data, our finding of an overall 9.5% higher mortality than expected across the 26 participating countries during the pandemic period aligns well with another recent study by M. Pizzato and colleagues, who estimated an 8.0% higher overall mortality than expected for the full four-year period 2020–2023 for 29 European countries[39]. However, a notable methodological difference in EuroMOMO's approach to estimation of expected levels of mortality is that only mortality data from spring and autumn weeks are included, as explained in the Supplementary Methods. Thereby, any excess mortality resulting from respiratory infections during winter seasons, e.g., influenza, or from heat waves during summer seasons, will not affect the expected mortality calculation, which is why this method ensures a better estimate of total excess mortality. Several other published studies on excess mortality during COVID-19[4,40] use a different approach, including seasonal periods of elevated mortality in the expected mortality calculation, hence introducing a bias towards an inflated baseline level and hence potentially underestimating the actual level of excess mortality in the population. The magnitude of difference in estimated excess mortality levels between these two approaches will however, require further in-depth comparative analysis to fully understand.

Weekly systematic mortality monitoring in a multi-country network serves as a simple yet powerful and sensitive means of detecting public health events leading to increased mortality. Excess mortality monitoring helps to inform clinicians, public health professionals, and the public, and guides countries in their epidemic preparedness and in providing a rapid and relevant public health response.

In conclusion, high levels of excess mortality were observed during the first and second wave of the COVID-19 pandemic in Europe,

 https://doi.org/10.1038/s41467-025-67981-1

particularly in older age groups, while excess mortality in the third and fourth wave were similar to previous influenza seasons. There were large inter-country differences in excess mortality. During the early stages of the COVID-19 pandemic, before implementation of control measures, the excess mortality in the population significantly exceeded the excess mortality associated with previous seasonal influenza epidemics.

Methods

Study design

We conducted a retrospective time-series analysis of excess all-cause mortality by age groups (0–14 years, 15–44 years, 45–64 years, 65–74 years, 75–84 years, and ≥ 85 years) estimated by the EuroMOMO network in the period from week 8/2020 to week 20/2023. For comparison, we estimated pooled excess all-cause mortality for periods of elevated mortality (excess ≥ 2 z-scores in three consecutive weeks) during the five most recent influenza seasons, i.e., defined as the winter periods from week 40 to week 20 in 2014/15, 2015/16, 2016/17, 2017/18, and 2018/19.

Participating countries

During the COVID-19 pandemic, 26 participating countries and subnational areas contributed with data to the pooled analyses: Austria, Belgium, Cyprus, Denmark, England (UK), Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Israel, Italy, Luxembourg, Malta, Netherlands, Norway, Northern Ireland (UK), Portugal, Scotland (UK), Slovenia, Spain, Sweden, Switzerland, and Wales (UK). See partner details on the EuroMOMO website (Euromomo.eu).

During previous influenza seasons from 2014/15 to 2018/19, 16 countries provided data systematically for all seasons, including the following countries: Belgium, Denmark, England (UK), Estonia, Finland, France, Greece, Hungary, Ireland, Netherlands, Norway, Portugal, Scotland (UK), Spain, Switzerland, and Wales (UK).

Collection of national all-cause mortality and population data

This study is based on the official national registration of deaths in the 26 participating countries. Each country has its own recording system, but all national analyses were conducted using the same EuroMOMO statistical algorithm using the software programs STATA or R. The results of the national analyses were submitted to the hub in either CSV or Excel format and collated in a database at Statens Serum Institut, Copenhagen. Information on race, ethnicity, or other socially relevant groupings were not relevant for this study.

For the COVID-19 pandemic period, i.e., week 8/2020, to week 20/2023, the country-specific numbers of all-cause deaths, aggregated by age groups and week of death, were compiled for analysis by week 26 in 2023.

For the previous influenza seasons, from week 40 to week 20, the country-specific mortality data for each season were compiled as follows: by week 28 in 2015, by week 28 in 2016, by week 28 in 2017, by week 28 in 2018, and by week 26 in 2019. Data was initially extracted by week 26, but if data from all countries was not available, the first subsequent week with complete data available was used.

Data on population size were extracted from Eurostat[41] for all countries, except Israel, for which the population was retrieved from the Health and Vital Statistics Sector, Central Bureau of Statistics, Israel. Population data were interpolated to weekly population sizes (ISOweeks) using linear interpolation based on the populations (as of January 1st) by country and age group. Person-years (pyrs) were defined as the sum of the observed person-time in the specific periods, to allow comparison of periods of different lengths.

Estimation of excess all-cause mortality

The number of excess all-cause deaths was calculated as the observed number of deaths subtracted from the expected number of deaths. The number of expected deaths is estimated based on the EuroMOMO methodology, see details in the Supplementary Methods. To avoid bias from the changed mortality pattern during the COVID-19 pandemic, data from 2020 to 2022 have been excluded from the estimations of expected mortality, such that the estimations are based on data from 2015 to 2019 and 2023.

Pooled excess mortality periods were defined as periods with weekly z-scores (a statistical measure that describes how many standard deviations the observed mortality differ from the expected) above two for a minimum of three consecutive weeks in pooled country data, which is the standard practice applied by EuroMOMO in the weekly mortality monitoring. These periods were defined based on pooled numbers of excess mortality, regardless of whether the individual countries observed increased mortality during that time, and used throughout the paper. See further details in the Supplementary Methods. The four longest periods of excess mortality were selected for a more detailed temporal and comparative analysis.

Estimates of excess mortality during the COVID-19 pandemic were calculated (a) weekly; (b) for the total period between week 8 in 2020 to week 20 in 2023, and (c) for each of the four defined periods of pooled excess mortality.

The excess mortality during the COVID-19 pandemic period was estimated at country level and pooled for all countries, for all ages, and stratified by age group, while the mortality estimated for the previous influenza seasons were limited to a pooled analysis for all ages and for those aged 65 years and above, due to implemented changes in definition of age group stratifications over time.

All-cause excess mortality rates were calculated by dividing the excess number of deaths with person-years (pyrs) observed over the period of interest. The percentage of all-cause excess number of deaths was calculated as the number of excess deaths divided by the expected number of deaths multiplied by 100%.

All estimates of excess deaths are presented as mortality rates with 95% confidence intervals. All analyses, figures, and tables were carried out using the statistical software R (version 4.3.1).

Ethics statement

As this was an observational, register-based study using only aggregated data with no individual-level information, formal ethics approval of the study protocol was not required. The study was approved by all participating national health authorities and endorsed by the European Centre for Disease Prevention and Control (ECDC) and the World Health Organization (WHO).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The national datasets underlying the findings of this study are not publicly available due to country-specific data protection regulations. The data are stored in controlled-access repositories at national institutes and may be obtained upon direct request to the respective national authorities (see https://euromomo.eu/about-us/partners for detailed contact information). However, updated national z-scores and aggregated data from the weekly EuroMOMO mortality monitoring are publicly available at from the EuroMOMO website: https://euromomo.eu/graphs-and-maps. Specific data inquiries may also be directed to the EuroMOMO hub at euromomo@ssi.dk. Source data are provided with this paper.

Code availability

The algorithm used nationally to calculate expected mortality and analyze the differences between observed and expected mortality per ISO week (as applied in the EuroMOMO network's weekly mortality monitoring) is available on GitHub: "JensXII/MOMO" (https://github.com/JensXII/MOMO).

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Acknowledgements
The "Excess mortality monitoring in Europe to assess impact of communicable diseases" is a project of the European Centre for Disease Prevention and Control (ECDC) run under the framework contract No. ECDC/2022/025. We thank ECDC and WHO for their continuous support.

Author contributions
S.K.N. and L.S.V. drafted the first version of the manuscript. S.K.N. performed the pooled analyses and provided the tables and figures. S.K.N., J.N., C.S., A.K., L.R., A.C., T.B., S.N.M., M.A., T.L., G.D., O.L., I.P., B.Z., M.A.H., K.M., I.P., A.P., T.M., E.K., N.R., F.K.D., C.D.B., J.M., A.V., K.E., N.C., L.A., J.K., C.M., S.S., A.P.R., N.K., I.L.G., D.G.B., I.G., A.F., R.W., N.A., T.C., M.B., D.T.B., N.W., M.H., P.J., R.P., N.B., B.S., T.G.K., K.M., and L.S.V. provided data, performed national analyses and/or contributed to the writing of the manuscript and approved the final version. The authors alone are responsible for the views presented in this manuscript and they do not necessarily reflect the views, decisions, or policies of the institutions with which the authors are affiliated.

Competing interests
The authors declare no competing interests.

Additional information
Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41467-025-67981-1.

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Peer review information *Nature Communications* thanks David Muscatello and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. A peer review file is available.

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